

classical era

foundation-model era

2019

2020

2021

2022

2023

2024

2025

2026

Embedding / Memory-bank

PaDiM

PatchCore

UniAD

SimpleNet

EfficientAD

Reconstruction / Synthesis

CutPaste

DRAEM

RD4AD

DiffusionAD

AnomalyDiffusion

RealNet

Anomaly Anything

Dinomaly

Anomaly Painter

MAGIC

CLIP / VFM zero/few-shot

WinCLIP

AnomalyCLIP

AnomalyDINO

FAPrompt

MuSc

AnomalyVFM

FoundAD

VisualAD

MLLM / Reasoning

GPT4V-AD

AnomalyGPT

MMAD

IAD-R1

AD-FM

JUDO

AnomalyClaw